

EXERCISES AND EXACT SOLUTIONS

Shelter Before Disaster

Independent practice, facilitated discussion, and assessment reference

Production Unit	PU-B07-C54
Notebook	V7_D_N04
Version	0.1
Assessment boundary	Learning evidence—not operational authorization

Instructions

Complete Part A before consulting the solutions. State assumptions explicitly. Where evidence is insufficient, write an abstention or escalation response rather than inventing precision.

Exercise 1

Add route/travel-time constraints.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 2

Model shelter failure.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 3

Add uncertainty in population estimates.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 4

Explain why the highest exposure zone may not receive the first intervention.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exact solutions and scoring guide

Solution 1

Use authoritative networks, hazards, modes, barriers, congestion, and accessibility.

Solution 2

Stress-test closure/capacity loss and identify redundancy.

Solution 3

propagate census/grid uncertainty through scenario totals and rank stability.

Solution 4

Urgency also depends on warning time, feasibility, vulnerability, access, cascading hazards, current evidence, and authorized response strategy.

Scoring rubric

Criterion	Strong evidence	Revision required
Decision framing	Owner, action, horizon, authority named	Model output substituted for decision
Evidence	Concepts, grain, provenance, limits explicit	Data treated as self-explanatory
Uncertainty	Abstention/escalation used appropriately	False precision or hidden limitations
Governance	Human authority and safeguards retained	Authority silently delegated